Euclid's Elements — Book I
Proposition 16

Formal Reconstruction — Technical Documentation

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The exterior-angle inequality for a triangle

1 Introduction

Euclid's Proposition I.16 is the exterior-angle inequality for a triangle. If a side of a triangle is extended beyond one of its vertices, the exterior angle formed at that vertex is greater than either of the two remote interior angles.

For a nondegenerate triangle ABC, extend BC through C to a point D, so that

$$B - C - D.$$

The proposition asserts

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD.$$

Unlike Proposition I.15, this proposition is not merely a thin application of one previously packaged theorem. Its reconstruction requires a substantial piece of Hilbert order-and-congruence geometry: an interior ray must be transported between congruent angles, and the resulting angle comparison must be expressed without numerical angle measure.

The formal development therefore isolates the geometric machinery in the Hilbert interface and leaves the Euclidean proposition itself comparatively short. In particular, the final Lean file contains three theorems:

- `euclid_proposition_16_first`,
- `euclid_proposition_16_second`,
- `euclid_proposition_16`.

2 The Classical Configuration

Let ABC be a nondegenerate triangle and let the side BC be produced through C to D:

$$B - C - D.$$

The angle $\angle ACD$ is exterior to the triangle. The two interior opposite angles are $\angle BAC$ and $\angle ABC$.

The formal hypotheses are

$$\neg\,\mathrm{Collinear}(A, B, C) \qquad \text{and} \qquad B - C - D.$$

The noncollinearity hypothesis makes the triangle nondegenerate, while strict betweenness records that D lies on the continuation of BC beyond C.

3 Statement

Euclid states Proposition I.16 as follows:

> If one side of a triangle is produced, the exterior angle is greater than either of the two interior opposite angles.

In the Hilbert reconstruction, strict comparison of angles is represented by `HilbertAngleLess`. Thus the formal conclusion is

$$\mathrm{HilbertAngleLess}(BAC, ACD) \quad \wedge \quad \mathrm{HilbertAngleLess}(ABC, ACD).$$

More explicitly,

$$\neg\,\mathrm{Collinear}(A, B, C), \quad B - C - D \Longrightarrow \begin{cases} \angle BAC < \angle ACD, \\ \angle ABC < \angle ACD. \end{cases}$$

The relation `HilbertAngleLess` is synthetic. It does not compare numerical measures. An angle is smaller than another when a congruent copy of the first can be realized as a proper interior subangle of the second.

4 Euclid's Classical Proof

Euclid proves the first inequality by a midpoint construction.

Let E be the midpoint of AC. Join BE and extend it beyond E to a point F such that

$$BE = EF.$$

Join CF.

Since $AE = EC$, $BE = EF$, and the vertical angles $\angle AEB$ and $\angle CEF$ are equal by Proposition I.15, the triangles AEB and CEF are congruent by Proposition I.4. Hence

$$\angle BAE = \angle ECF.$$

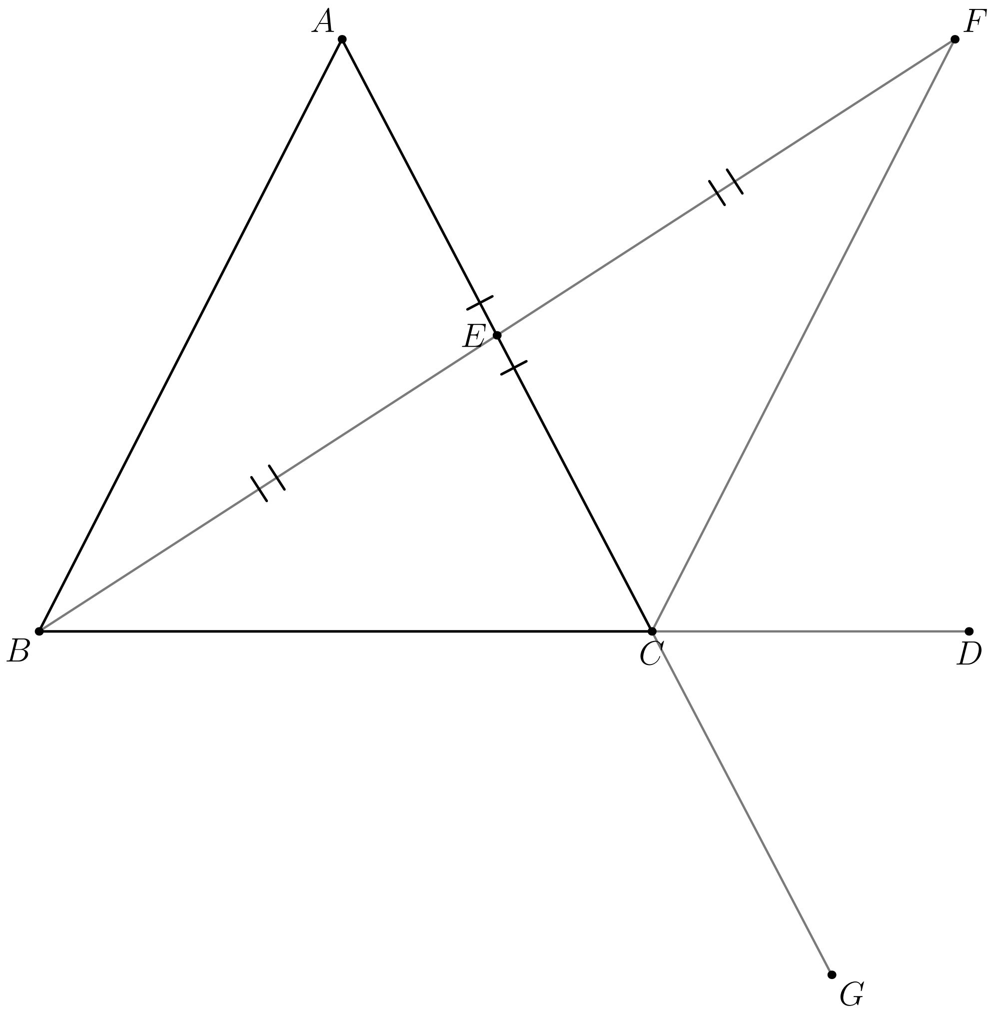

Figure 1: Classical Wilkins configuration for the first inequality. The midpoint construction uses $A - E - C$, $AE \cong EC$; BE is produced through E to F with $BE \cong EF$; BC is produced through C to D, and AC through C to G. The ray CF lies in the exterior angle only after the required side/order argument.

Since E lies on AC, this gives

$$\angle BAC = \angle ACF.$$

But the ray CF lies inside the exterior angle ACD. Therefore

$$\angle BAC < \angle ACD.$$

For the second opposite angle Euclid repeats the same argument after extending another side of the triangle. The equality of the corresponding vertical exterior angles then identifies the resulting exterior angle with $\angle ACD$.

Thus the classical architecture is

$$\text{midpoint construction} \to \text{I.15} \to \text{SAS (I.4)} \to \text{interior subangle} \to \text{I.16}.$$

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction retains the synthetic structure of Euclid's argument but packages the difficult order-theoretic steps into reusable interface theorems.

For the first opposite angle the auxiliary construction produces points M and E with

$$A - M - C, \qquad AM \cong MC,$$

$$B - M - E, \qquad BM \cong ME,$$

and

$$\angle BAC \cong \angle MCE.$$

These are the exact geometric outputs unpacked from `hilbert_exterior_angle_aux`. From these data the interface proves that AB is parallel to CE. The theorem `hilbert_exterior_angle_less` then concludes directly that

$$\angle BAC < \angle ACD.$$

Schematically:

$$\begin{array}{c} \neg\,\text{Collinear}(A, B, C) \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_aux} \\ \downarrow \\ A - M - C,\ B - M - E,\ AM \cong MC,\ BM \cong ME,\ \angle BAC \cong \angle MCE \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_aux\_parallel} \\ \downarrow \\ AB \parallel CE \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_less} \\ \downarrow \\ \angle BAC < \angle ACD. \end{array}$$

The second inequality is deliberately reduced to the first. Extend AC through C to F. Apply the first part to triangle BAC to obtain

$$\angle ABC < \angle BCF.$$

The lines ACF and BCD intersect at C. Classically this is the vertical-angle content of Proposition I.15; the current Lean proof calls the generic theorem `VerticalAngles` directly and obtains, after symmetry and ray reversal,

$$\angle BCF \cong \angle ACD.$$

Transporting the right-hand side of the strict angle comparison across this congruence yields

$$\angle ABC < \angle ACD.$$

Thus the second half has the compact architecture

first half of I.16 $\to$ `VerticalAngles` $\to$ transport of $<\to$ second half of I.16.

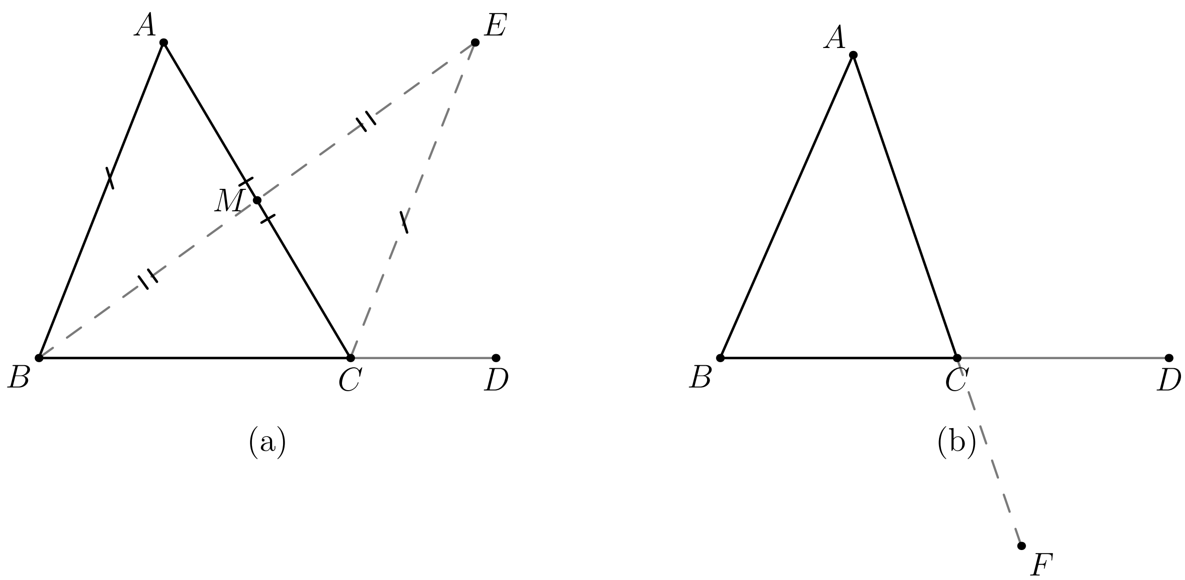

Figure 2: Formal Lean geometry. (a) The first branch unpacks M, E with $A-M-C$, $AM \cong MC$, $B-M-E$, $BM \cong ME$; the same helper data yield $\angle BAC \cong \angle MCE$ and then $AB \parallel CE$. (b) For the second branch Lean constructs $A-C-F$; together with $D-C-B$ this gives the vertical-angle configuration $\angle BCF \cong \angle ACD$.

6 Hilbert Reconstruction

6.1 The First Opposite Interior Angle

The theorem `euclid_proposition_16_first` begins from

$$\neg\,\mathrm{Collinear}(A, B, C), \qquad B-C-D.$$

It invokes

```
hilbert_exterior_angle_aux
```

to obtain points M, E and the auxiliary configuration required for the exterior-angle argument.

The same configuration is passed to

`hilbert_exterior_angle_aux_parallel`,

which proves

$$AB \parallel CE.$$

Finally,

`hilbert_exterior_angle_less`

combines the auxiliary construction, the extension $B - C - D$, and the parallelism to establish

$$\angle BAC < \angle ACD.$$

The Euclidean theorem itself therefore contains no unresolved construction. The substantial work has been moved into the Hilbert interface, where it can be reused independently of Proposition I.16.

6.2 The Second Opposite Interior Angle

For the second inequality, the proof first derives $A \neq C$ from the noncollinearity of A, B, C. Hilbert's order axiom then extends AC through C to a point F:

$$A - C - F.$$

The triangle is viewed in the order BAC. Applying `euclid_proposition_16_first` to this reordered triangle gives

$$\angle ABC < \angle BCF.$$

It remains to identify $\angle BCF$ with the desired exterior angle $\angle ACD$. From $B - C - D$ we obtain the reversed betweenness relation

$$D - C - B.$$

The two straight lines ACF and BCD therefore form a pair of vertical angles at C. The current Lean proof calls `VerticalAngles` directly (the generic theorem corresponding to the classical content of I.15) and, after the required symmetry and reversal operations, obtains

$$\angle BCF \cong \angle ACD.$$

The theorem

`hilbert_angleLess_transport_right`

then transports the comparison across the congruent right-hand angle:

$$\angle ABC < \angle BCF, \qquad \angle BCF \cong \angle ACD \quad \Longrightarrow \quad \angle ABC < \angle ACD.$$

7 Lean Formalization

The first half is formalized as follows:

```
theorem euclid_proposition_16_first
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D := by

  rcases
      hilbert_exterior_angle_aux
        Geo A B C hABC with
    <M, E,
      hAMC,
      hAMMC,
      hBME,
      hBMME,
      hAngle>

  have hParallel :
      Geo.Parallel A B C E :=
    hilbert_exterior_angle_aux_parallel
      Geo A B C M E
      hABC
      hAMC
      hBME
      hAngle

  exact
    hilbert_exterior_angle_less
      Geo A B C D M E
      hABC
      hAMC
      hBME
      hBCD
      hAngle
      hParallel
```

The second half deliberately reuses the first:

```
have hLessBCF :
    HilbertAngleLess Geo A B C B C F :=
  euclid_proposition_16_first
    Geo
    B A C F
    hBACnc
    hACF
```

The vertical-angle relation is then constructed at C and normalized to the required orientation:

```
have hVerticalRaw :
    Geo.AngleCongruent A C D F C B :=
  VerticalAngles
```

```
    Geo
    A C D F B
    hACF
    hDCA
    hACDnc

have hVerticalSymm :
    Geo.AngleCongruent F C B A C D :=
  Geo.angle_congruent_symmetry
    A C D
    F C B
    hVerticalRaw

have hVertical :
    Geo.AngleCongruent B C F A C D :=
  (Geo.angle_congruent_reverse_first
    F C B A C D).mp
    hVerticalSymm
```

Finally the strict inequality is transported:

```
exact
  hilbert_angleLess_transport_right
    Geo
    A B C
    B C F
    A C D
    hLessBCF
    hACDnc
    hVertical
```

The complete proposition simply combines the two parts:

```
theorem euclid_proposition_16
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D /\
    HilbertAngleLess Geo A B C A C D := by

  have hFirst :=
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

  have hSecond :=
    euclid_proposition_16_second
      Geo A B C D hABC hBCD

  exact <hFirst, hSecond>
```

8 Relationship with Earlier Propositions

8.1 Proposition I.10

Euclid's classical proof uses a midpoint construction as part of the auxiliary configuration. Proposition I.10 is therefore one of the constructional ingredients behind the traditional argument. In the present Lean file, however, no call to `euclid_proposition_10` occurs: the midpoint-style geometry has already been packaged inside the Hilbert interface helper `hilbert_exterior_angle_aux`.

8.2 Proposition I.15

Classically, vertical-angle equality is Proposition I.15. The second formal branch uses exactly that geometry after constructing $A - C - F$ and reversing $B - C - D$ to $D - C - B$. The current production file does not import `Proposition15` or call `euclid_proposition_15`; it calls the generic interface theorem `VerticalAngles` directly. Thus I.15 is the classical correspondence, while `VerticalAngles` is the direct formal dependency.

8.3 Transition to Proposition I.17

Proposition I.16 supplies the strict exterior-angle comparisons used immediately in I.17. Once a side is extended, I.17 can represent "two angles together less than two right angles" by comparing an interior angle with the supplementary exterior angle. I.16 therefore begins the strict angle-order block that continues through I.17–I.19.

9 Hilbert and Book Zero Components Used

9.1 `HilbertAngleLess`

The proposition requires a genuinely synthetic notion of strict angle comparison. The relation `HilbertAngleLess` expresses that a congruent copy of one angle occurs as a proper interior subangle of another. This avoids introducing numerical angle measure and keeps Proposition I.16 inside pure incidence, order, and congruence geometry.

9.2 Auxiliary Exterior-Angle Construction

This theorem supplies the auxiliary points used in the first exterior-angle construction. It packages the midpoint-style configuration that underlies Euclid's original proof and returns the betweenness and congruence data required by the subsequent argument.

9.3 Derived Parallelism

From the auxiliary configuration, this theorem proves

$$AB \parallel CE.$$

The parallel relation is not an additional Euclidean postulate here; it is a derived relation inside the neutral Hilbert development used to organize the exterior-angle argument.

9.4 Exterior-Angle Strict Comparison

This is the principal interface theorem used by the first half of I.16. It converts the auxiliary construction and its parallelism into the strict angle comparison

$$\angle BAC < \angle ACD.$$

Its proof depends on the interior-subangle machinery developed in the Hilbert layer.

9.5 Interior-subangle transport

A substantial part of the work required for Proposition I.16 lies below the final Euclidean file. The Hilbert interface establishes that an interior ray can be transported between congruent angles while preserving the corresponding subangle.

The completed infrastructure includes the segment and crossbar mechanisms needed for that transport, in particular the ability to transport strict segment comparison through congruence, transport an interior point between congruent segments, and preserve ray–segment intersection when the endpoints are moved along the same two rays.

This is the part of the reconstruction where the formal proof exposes geometric structure that is largely implicit in the paper proof.

9.6 `VerticalAngles`

The second half uses the generic theorem `VerticalAngles`, which is the formal interface counterpart of the vertical-angle content of Proposition I.15. After AC is extended through C to F, the angles

$$\angle BCF \quad \text{and} \quad \angle ACD$$

are vertical. Their congruence allows the second inequality to be reduced to the first.

9.7 Right-Hand Angle Transport

This theorem expresses invariance of strict angle comparison under congruence of the right-hand angle. In the final step,

$$\angle ABC < \angle BCF$$

and

$$\angle BCF \cong \angle ACD$$

imply

$$\angle ABC < \angle ACD.$$

10 Formalization Notes

Proposition I.16 is the transition from angle congruence to strict angle comparison. The production theorem uses `HilbertAngleLess`; no numerical angle measure is introduced. The first inequality is delegated to reusable exterior-angle infrastructure, while the second reuses the first after a genuinely new extension $A - C - F$ and a vertical-angle transport.

The two branches are not geometric cases: both conclusions hold simultaneously. Their asymmetry is representational and proof-theoretic. The first branch exposes the hidden midpoint-style configuration M, E; the second creates a new extension and normalizes the resulting vertical angle to the statement orientation.

11 Diagrammatic Audit

The preserved Book1R chapter contained two figures: a statement-only extension $B - C - D$ and the classical Wilkins construction. The current production Lean reveals two additional proposition-level geometric states. The audited figure set therefore records the geometry as follows.

- Figure 1 preserves the classical Wilkins construction. It is kept separate from the formal helper because the point names and proof architecture are not the same: classical E, F correspond qualitatively to formal M, E, but the two layers must not be conflated.
- Figure 2(a) records the exact output of `hilbert_exterior_angle_aux`: $A - M - C$, $AM \cong MC$, $B - M - E$, $BM \cong ME$, together with the auxiliary ray CE. The derived relation $AB \parallel CE$ is marked on the same panel because it adds no new points or incidence state.
- Figure 2(b) records the genuinely new extension $A - C - F$ used in `euclid_proposition_16_second`. Together with $D - C - B$ it forms the vertical-angle configuration at C.
- The former statement-only figure $B - C - D$ is not retained as a separate float: that geometry is present in both substantive figures. Likewise, congruence symmetry, angle reversal, and `hilbert_angleLess_transport_right` change the logical reading of an existing configuration rather than creating new geometry.
- Internal witnesses used below `hilbert_exterior_angle_less` or inside the representation of `HilbertAngleLess` are reusable interface witnesses. They are described in the dependency discussion but are not proposition-specific diagram states.

Thus the audited chapter uses two figures with three panels: one classical panel and a two-panel formal figure. A figure records geometry, not tactic state.

12 Dependency Summary

The formal dependency structure can be summarized as

$$\begin{array}{c}
\text{Hilbert incidence} + \text{order} + \text{congruence} \\
\downarrow \\
\text{segment construction and ray/order theory} \\
\downarrow \\
\text{interior-subangle transport / crossbar machinery} \\
\downarrow \\
\texttt{hilbert_exterior_angle_less} \\
\downarrow \\
\texttt{euclid_proposition_16_first} \\
\downarrow \\
\texttt{VerticalAngles} + \text{ angle-less transport} \\
\downarrow \\
\texttt{euclid_proposition_16_second} \\
\downarrow \\
\texttt{euclid_proposition_16}.
\end{array}$$

The final Euclidean theorem is therefore compact, but its compactness is meaningful: the difficult geometry has been factored into reusable Hilbert-interface results rather than hidden in an axiom or a local `sorry`.

13 Conclusion

Proposition I.16 marks a significant step beyond the immediately preceding propositions. It introduces a strict comparison of angles and requires a robust synthetic notion of interiority. In Euclid's presentation this is handled by a midpoint construction, SAS, and the observation that one angle is a proper part of another. In the formal Hilbert reconstruction the same content is decomposed into explicit reusable mechanisms for rays, betweenness, congruence, crossbar behavior, and transport of strict angle comparison.

The final Lean theorem is short precisely because those mechanisms have been established independently. The current `Proposition16.lean` contains no local `sorry` or local axiom declaration. This documentation audit did not run `lake build` or `#print axioms`, so no fresh compilation or axiomatic-status claim is made here.